YANG XU

Curriculum Vitae

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Department of Computer Science and Engineering
Southern University of Science and Technology

RESEARCH INTERESTS

Yang Xu's research interest lies in the intersection of computational linguistics, natural language processing, psycholinguistics, and cognitive sciences. He develops theoretical paths and computational methods for better understand human language, cognition, and the mysterious consciousness of mind. His research draws inspirations from Information-Theory, theories of mind (in psychology). He directs the Computational Linguistics and Consciousness Science (CLCS) Lab in SUSTech, a group of researchers enthusiastic in the following research questions:

- How natural languages differ from those generated by large language models (LLMs)?
- How the information in different communication modalities (verbal and non-verbal) distribute and evolve in monologue/dialogue?
- How the dyad of speakers adjust their language to reach better coordination in dialogue and to maintain efficient communication as a dynamical system?
- Connection between language production and the cognitive/physiological constraints.

PROFESSIONAL EXPERIENCE

Associate Professor June 2023 – Present
Department of Computer Science and Engineering
Southern University of Science and Technology (SUSTech), Shenzhen, China

Assistant Professor Aug 2018 – May 2023
Department of Computer Science, San Diego State University (SDSU), CA, US

EDUCATION

Ph.D in Information Sciences and Technology Aug 2013 – May 2018
The Pennsylvania State University, USA

Master of Art Sep 2010 – July 2013
Department of Psychology, Tsinghua University, China

Bachelor of Engineering Sep 2006 – July 2010
Department of Electronic Engineering, Tsinghua University, China

RESEARCH GRANTS

PI, General Program of Shenzhen Natural Science Foundation, 2024–2027
Research on Hallucination in Large Language Models Based on Information Entropy Theory

PI, SUSTech–Huawei Trusted Intelligent Systems Innovation Laboratory (Completed), 2025
Entropy-based Evaluation of Large Language Model Generation

PI, NSF-CRII-HCC (Award ID: 2105192), $166K, 2021 – 2024
Modeling Computer-Mediated Task-Oriented Dialogues with Multi-Modality Information Theoretic

Approaches

Co-PI, NIH (subaward to SDSU HealthLink), $50K, 2021 – 2022
An NLP Approach to Analyze the Past and Present, and Inform the Future of Health Disparities Research

PUBLICATIONS

1. Zhichen Liu, Tianle Lun, Zhibin Wen, Hao An, Yulin Ou, Jianhui Xu, Hao Zhang, Wenyi Fang, Yang Zheng, and **Yang Xu**. Fast and accurate probing of in-training LLMs' downstream performances. In *Advances in Neural Information Processing Systems (NeurIPS 2026)*, 2026. (CCF-A)
2. Yongyuan Li and **Yang Xu**. Uniform information density-based preference optimization. In *Conference on Language Modeling (COLM 2026)*, 2026
3. Hao An and **Yang Xu**. Teaching LLMs to abstain via fine-grained semantic confidence reward. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (Main Conference)*, 2026. (CCF-B)
4. Yu Wang, Leyi Lao, Langchu Huang, Gabriel Skantze, **Yang Xu**, and Hendrik Buschmeier. Investigating the representation of backchannels and fillers in fine-tuned language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5319–5348, San Diego, California, United States, July 2026. Association for Computational Linguistics. (Oral; CCF-A)
5. Zhichen Liu, Yongyuan Li, and **Yang Xu**. Think in sentences: Explicit sentence boundaries enhance language model's capabilities. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2275–2288, San Diego, California, United States, July 2026. Association for Computational Linguistics. (Oral; CCF-A)
6. Yulin Ou, Yu Wang, **Yang Xu**, and Hendrik Buschmeier. Identifying the periodicity of information in natural language. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1161–1175, San Diego, California, United States, July 2026. Association for Computational Linguistics. (CCF-A)
7. Zhongyu Yang, Dannong Xu, Yonghan Zhang, Kefan Chen, Xinyi Wang, **Yang Xu**, Wei Pang, and Yingfang Yuan. Do vision and text cues exhibit evidential coupling? UFO: A benchmark for compositional multimodal reasoning in unified models. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, 2026. (CCF-A)
8. Hao An, Yibin Lou, Jiayi Guo, and **Yang Xu**. Purging the gray zone: Latent-geometric denoising for precise knowledge boundary awareness. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 2562–2576, San Diego, California, United States, July 2026. Association for Computational Linguistics. (CCF-A)
9. Zhichen Liu, Kaitong Qin, Linhan He, and **Yang Xu**. WaveDetect: Robust framework for machine-generated text detection via wavelet transform. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 8712–8727, San Diego, California, United States, July 2026. Association for Computational Linguistics. (CCF-A)
10. Jianjie Zheng, Zhichen Liu, Zhanyu Shen, Jingxiang Qu, Guanhua Chen, Yile Wang, **Yang Xu**, Yang Liu, and Sijie Cheng. Evaluating memory capability in continuous lifelog scenario. In *Findings of the Association for Computational Linguistics: ACL 2026*, 2026. (CCF-A)

11. Xiaoshan He, Xiaoqun Liu, Haodong He, Yu Wang, and **Yang Xu**. Eye movement features can predict human preferences on machine-generated texts. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 470–486, San Diego, California, United States, July 2026. Association for Computational Linguistics

12. Zhichen Liu, Yongyuan Li, **Yang Xu**, Yu Wang, Yingfang Yuan, and Zuhao Yang. Evaluating text generation quality using spectral distances of surprisal. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025

13. Zhongyu Yang, Junhao Song, Zhang Luo, Zuhao Yang, **Yang Xu**, Jingfen Lan, Yonghan Zhang, Wei Pang, Siyang Song, and Yingfang Yuan. Rechar: Revitalising characters with structure preserved and user-specified aesthetic enhancements. In *Proceedings of the SIGGRAPH Asia 2025 Technical Communications*, SA Technical Communications '25. Association for Computing Machinery, 2025

14. Lam Nguyen and **Yang Xu**. Reasoning for translation: Comparative analysis of chain-of-thought and tree-of-thought prompting for LLM translation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 259–275, Vienna, Austria, July 2025. Association for Computational Linguistics

15. **Yang Xu**, Yu Wang, Hao An, Zhichen Liu, and Yongyuan Li. Detecting subtle differences between human and model languages using spectrum of relative likelihood. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 10108–10121, Miami, Florida, USA, November 2024. Association for Computational Linguistics. (CCF-B)

16. Yu Wang, **Yang Xu**, Gabriel Skantze, and Hendrik Buschmeier. How much does nonverbal communication conform to entropy rate constancy?: A case study on listener gaze in interaction. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 3533–3545. Association for Computational Linguistics, 2024. (CCF-A)

17. Zuhao Yang, Yingfang Yuan, **Yang Xu**, Shuo Zhan, Huajun Bai, and Kefan Chen. Face: Evaluating natural language generation with fourier analysis of cross-entropy. *Advances in Neural Information Processing Systems*, 36, 2024. (CCF-A, co-first author)

18. **Yang Xu** and Yang Cheng. Spontaneous gestures encoded by hand positions improve language models: An information-theoretic motivated study. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 9409–9424, 2023. (CCF-A)

19. **Yang Xu**, Yang Cheng, and Riya Bhatia. Gestures are used rationally: Information theoretic evidence from neural sequential models. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 134–140, Gyeongju, Republic of Korea, 2022. International Committee on Computational Linguistics. (CCF-B)

20. Bruce I. Gaynes, Mark B. Shapiro, Jie Xu, Abel Saju Augustine, **Yang Xu**, and et.al. Hierarchical data visualization of experimental erythrocyte aggregation employing cross correlation and optical flow applications. *Microvascular Research*, 143:104386, 2022

21. **Yang Xu**. Global divergence and local convergence of utterance semantic representations in dialogue. In *Proceedings of the Society for Computation in Linguistics*, volume 4, pages 116–124. 2021

22. **Yang Xu** and Zheng-sheng Zhang. Historical changes in semantic weights of sub-word units. In Adam Jatowt, Nina Tahmasebi, Yang Xu, and Lars Borin, editors, *Computational Approaches to Semantic Change*, pages 169–187. Language Science Press, 2021

23. M Alex Kelly, **Yang Xu**, Jesús Calvillo, and David Reitter. Which sentence embeddings and which layers encode syntactic structure? In *Proceedings of the 42nd Annual Meeting of the Cognitive Science Society*, pages 2375–2381, 2020. (acceptance rate: 18.1%; CCF-B)

24. **Yang Xu**, Jiasheng Zhang, and David Reitter. Treat the word as a whole or look inside? subword

embeddings model language change and typology. In *Proceedings of the 1st International Workshop on Computational Approaches to Historical Language Change*, pages 136–145, Florence, Italy, 2019. Association for Computational Linguistics

25. **Yang Xu**, Jeremy Cole, and David Reitter. Linguistic alignment is affected more by lexical surprisal rather than social power. *Proceedings of the Society for Computation in Linguistics*, 2(1):349–352, 2019

26. **Yang Xu**, Jeremy Cole, and David Reitter. Not that much power: Linguistic alignment is influenced more by low-level linguistic features rather than social power. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, volume 1, pages 601–610, 2018. (acceptance rate: 24.5%; CCF-A)

27. **Yang Xu** and David Reitter. Information density converges in dialogue: Towards an information-theoretic model. *Cognition*, 170:147–163, 2018. (Impact factor: 3.65; CCF-B)

28. **Yang Xu** and David Reitter. Spectral analysis of information density in dialogue predicts collaborative task performance. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 623–633, Vancouver, Canada, 2017. (acceptance rate: 25%; CCF-A)

29. **Yang Xu** and David Reitter. Entropy converges between dialogue participants: Explanations from an information-theoretic perspective. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 537–546, Berlin, Germany, August 2016. Association for Computational Linguistics. (acceptance rate: 28.0%; CCF-A)

30. **Yang Xu** and David Reitter. Convergence of syntactic complexity in conversation. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 443–448, Berlin, Germany, August 2016. Association for Computational Linguistics. (acceptance rate: 28.0%; CCF-A)

31. **Yang Xu** and David Reitter. An evaluation and comparison of linguistic alignment measures. In *Proceedings of Cognitive Modeling and Computational Linguistics (CMCL)*, pages 58–67, Denver, CO, 2015. Association for Computational Linguistics

32. Alexander G. Ororbia II, **Yang Xu**, Vito D'Orazio, and David Reitter. Error-correction and aggregation in crowd-sourcing of geopolitical incident information. In N. Agarwal et al., editor, *Social Computing, Behavioral Modeling and Prediction*, volume 9021 of *Lecture Notes in Computer Science*, pages 381–387. Springer, 2015

33. David Reitter, **Yang Xu**, Patrick Craven, Anikó Sándor, R. Chris Garrett, E. Vince Cross, and Jerry L. Franke. Cognitive models predicting surprise in robot operators. In *Proc. International Conference on Cognitive Modeling*, pages 190–191, Groningen, Netherlands, 2015

34. **Yang Xu** and Hong Li. Influence of visibility range and degree of urgency on efficiency of evacuation: Mediating effect of herding behavior. *Studies of Psychology and Behavior*, 13(3):311–319, 2015. (Published in Chinese)

35. Hong Li, **Yang Xu**, Shi Chen, and Anqi Gao. Effects of intuition and analysis on the adoption of sources of information in escape decision-making. *Studies of Psychology and Behavior*, 10(6):452–458, 2012. (Published in Chinese)

CONFERENCE PRESENTATIONS (SELECTED)

1. Global Divergence and Local Convergence of Dialogue, virtual, *SCiL*, 2021
2. Subword Embeddings Reveal Language Change, poster, *CogSci*, Montreal, Canada, 2019
3. Treat the Word As a Whole or Look Inside, poster, *ACL*, Florence, Italy, 2019

INVITED TALKS

University of Waterloo, Canada — Aug 2022
Host: Dr. Moojan Ghafurian
Title: Speaking Rationally by Gestures: Evidence from Multi-Modal Language Models

Beijing Language and Culture University — Aug 2021
Host: Dr. Qun Guan
Title: The Information-Theoretic Insights of Multimodal Language Models

Beijing Language and Culture University — Aug 2020
Host: Dr. Qun Guan
Title: Entropy Constancy: Insights about Natural Language from the Perspective of Information

Westlake University, China — Nov 2019
Host: NLP Group of Dr. Yue Zhang *Title*: Dialogue from an Information-Theoretic Perspective

Shanghai Jiao-Tong University, China — Nov 2019
Host: College of Engineering *Title*: Efficient Communication in Dialogue

Tsinghua University, China — Nov 2019
Host: Dept. of Psychology *Title*: An Information Theoretic Psycholinguistic Model of Dialogue

San Diego State University — Oct 2019
Host: Dept. of Math and Stats *Title*: Statistical Methods Used in Natural Language Processing

San Diego State University — Oct 2019
Host: SDSU A.I. Seminar *Title*: Recent Advances in Natural Language Processing

San Diego State University — Feb 2018
Host: Dept. of CS *Title*: Computational Understanding of Dialogue and Language

HONOR AND AWARDS

Award for Research Excellence — Penn State, College of Information Sciences and Technology, 2018
Outstanding Graduates Award — Tsinghua University, 2013
Distinguished Master Thesis Award — Tsinghua University, 2013
Fellowship of Excellence in College Entrance Exam — Changxing, Zhejiang, 2006

PROFESSIONAL SERVICES

Area Chair, ACL 2025–2026
Reviewer, CogSci 2018–2020
Reviewer, *Lingua* and *Dialogue and Discourse*, 2022

COURSES TAUGHT

CS460 Algorithms — 2018 – 2022

CS549 Machine Learning — 2018 – 2022